Chapter 13. Classification of Covering Spaces

Equivalence of coverings, universal covering space, covering transformations, existence of coverings

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Throughout, a covering map $p : E \to B$ has E, B path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0)) \cong \pi_1(E, e_0)$ (Theorem 9.17).

13.1 Equivalence of covering spaces

Motivation. Covering spaces are classified by subgroups of π_1, as in Galois theory.

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

Lemma 13.1 (79.1, general lifting lemma)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where Y is path connected and locally path connected. Then f has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

Proof. If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$.

Uniqueness: for a path α from y_0 to y_1, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from e_0, so $\tilde{f}(y_1)$ is its end point.

Existence: let $\tilde{f}(y_1)$ be the end point of the lifting of $f \circ \alpha$ from e_0. Well defined: for another path β, lift $f \circ \alpha$ to γ from e_0 and $f \circ \bar{\beta}$ to δ from $\gamma(1)$; $\gamma * \delta$ lifts $f \circ (\alpha * \bar{\beta})$, whose class is in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); thus $\bar{\delta}$ lifts $f \circ \beta$ from e_0 and ends at $\gamma(1)$. Continuity at y_1: given $N \ni \tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice V_0 through $\tilde{f}(y_1)$ lies in N, and a path-connected $W \ni y_1$ with $f(W) \subset U$. For a path δ in W from y_1 to y, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in V_0. So $\tilde{f}(W) \subset V_0 \subset N$. ∎

Theorem 13.2 (79.2)

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such h is unique.

Proof. If h exists, h_* is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of p, $h(e_0) = e_0'$, and $k : E' \to E$ of p', $k(e_0') = e_0$. $k \circ h$ and id_E both lift p and fix e_0, so are equal; likewise $h \circ k = \mathrm{id}$. Uniqueness of h is that of liftings. ∎

Lemma 13.3 (79.3)

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$.
(a) If γ is a path in E from e_0 to e_1 and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$.
(b) Conversely, if H is conjugate to H_0, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

Proof. (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_1; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at e_0 over $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$.

(b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift α to γ from e_0, $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ∎

Theorem 13.4 (79.4)

p and p' are equivalent iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. If h is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ (Theorem 13.2), conjugate to H_0' (Lemma 13.3). Conversely, Lemma 13.3(b) gives e_1' with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 an equivalence $e_0 \mapsto e_1'$. ■

Example 13.1

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$, so every covering of S^1 is equivalent to $\mathbb{R} \to S^1$ (trivial subgroup) or $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

13.2 The universal covering space

Motivation. The simply connected covering lies above all others, like the splitting field.

If E is simply connected, p is a **universal covering**; it is unique up to equivalence (Theorem 13.4).

Lemma 13.5 (80.1)

Let B be path connected and locally path connected, and $p : E \to B$ a covering map with E not necessarily path connected. If E_0 is a path component of E, then $p|E_0 : E_0 \to B$ is a covering map.

Proof. E is locally path connected, so E_0 and $p(E_0)$ are open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice V meets E_0, hence lies in E_0, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$, and $(p|E_0)^{-1}(U)$ is the union of the slices meeting E_0, each in E_0 and mapped homeomorphically onto U. ■

Lemma 13.6 (80.2)

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected).
(a) If p and r are covering maps, so is q.
(b) If p and q are covering maps, so is r.

Proof. Let $x_0 \in X$, $y_0 = q(x_0)$.

(a) q is onto: for a path $\tilde{\alpha}$ in Y from y_0 to y, lift $r \circ \tilde{\alpha}$ through p to $\tilde{\tilde{\alpha}}$ from x_0; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through r from y_0, so equals $\tilde{\alpha}$ and ends at y. Given y, take a path-connected $U \ni r(y)$ evenly covered by p and r, and the slice $V \ni y$ of $r^{-1}(U)$. Each slice U_α of $p^{-1}(U)$ is connected, so q maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those U_α mapped into V, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto V.

(b) Let $U \ni z$ be path connected and evenly covered by p, and V_β the path components of $r^{-1}(U)$, open and disjoint. The slices U_α of $p^{-1}(U)$ are its path components, and q maps each into one V_β; so $q^{-1}(V_\beta)$ is a union of slices containing some U_α (q onto). By Theorem 9.10, $q^{-1}(V_\beta) \to V_\beta$ is a covering map with path component U_α, so $q|U_\alpha : U_\alpha \to V_\beta$ is onto (Lemma 13.5); it is open and injective (as $p|U_\alpha$ is), hence a homeomorphism, and $r|V_\beta = (p|U_\alpha) \circ (q|U_\alpha)^{-1}$ is a homeomorphism onto U. ■

Theorem 13.7 (80.3)

If $p : E \to B$ is a covering map with E simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

Proof. Lemma 13.1 applies trivially and gives q with $r \circ q = p$; Lemma 13.6(a) shows q is a covering map. ■

Lemma 13.8 (80.4)

If $p : E \to B$ is a covering map with E simply connected and $p(e_0) = b_0$, then b_0 has a neighbourhood U such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

Proof. Let U be evenly covered, $\tilde{U}$ the slice containing e_0. A loop f in U lifts to a loop in $\tilde{U}$, nulhomotopic in E; applying p gives a path homotopy from f to a constant. ■

Example 13.2

The infinite earring X (Example 11.2) is path connected and locally path connected but has no universal covering: every neighbourhood U of the origin contains some C_n, a retract of X, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

13.3 Covering transformations

Motivation. Deck transformations, the analogue of the Galois group, give $\pi_1(B)$ for the universal cover.

An equivalence of p with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of H in G is $N(H) = \{g : gHg^{-1} = H\}$. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection from the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

Lemma 13.9 (81.1)

The image of Ψ equals $\Phi(N(H_0)/H_0)$.

Proof. Let γ lift a loop α from e_0, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. Some h has $h(e_0) = e_1$ iff $H_1 = H_0$ (Theorem 13.2); as $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), iff $[\alpha] \in N(H_0)$. ■

Theorem 13.10 (81.2)

$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

Proof. It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, γ, δ paths from e_0 to e_1, e_2, $\alpha = p \circ \gamma$, $\beta = p \circ \delta$, so $\Phi([\alpha]H_0) = e_1$, $\Phi([\beta]H_0) = e_2$. $h \circ \delta$ lifts β from e_1 to $e_3 = h(k(e_0))$, so $\gamma * (h \circ \delta)$ lifts $\alpha * \beta$ from e_0 to e_3: $\Phi([\alpha * \beta]H_0) = \Psi(h \circ k)$. ■

Corollary 13.11 (81.3)

H_0 is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries e_1 to e_2. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

Proof. If H_0 is normal, Ψ is onto, giving $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes Ψ onto, so $N(H_0) = \pi_1(B, b_0)$. ■

Corollary 13.12 (81.4)

If E is simply connected, $\mathcal{C}(E,p,B) \cong \pi_1(B,b_0)$.

If H_0 is normal, p is **regular**, and $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path β from b to b_0, lift β from x_1, x_2 to paths ending at e_1, e_2, and let $h(e_1) = e_2$; h carries the first lifting to a lifting of β ending at e_2, by uniqueness (lift $\bar{\beta}$ from e_2) the second, so $h(x_1) = x_2$.

Example 13.3

Every covering of S^1 is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around A lifts to a non-loop from the origin but to a loop from every other fibre point on the y-axis, similarly for B, so every covering transformation fixes the origin.

For a group G of homeomorphisms of X, the **orbit space** X/G is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each x has a neighbourhood U with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

Theorem 13.13 (81.5)

Let X be path connected and locally path connected and G a group of homeomorphisms of X. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case π is regular and G is its group of covering transformations.

Proof. π is open: $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous, $\pi^{-1}\pi(U)$ is the disjoint union of the open $g(U)$, each meeting each orbit at most once; $\pi|g(U)$ is a continuous open bijection onto $\pi(U)$, a homeomorphism. Conversely, if π is a covering, let U be the slice through x over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u, g(u) \in U$, then $g(u) = u$ ($\pi|U$ injective), so g is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each g is a covering transformation; if h is one with $h(x_1) = x_2$, then $g(x_1) = x_2$ for some g, and $h = g$ (Theorem 13.2). G is transitive on fibres, so π is regular (Corollary 13.11). ■

Theorem 13.14 (81.6)

If $p : X \to B$ is regular with group of covering transformations G, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

Proof. p is constant on orbits, so induces a continuous k. p is a quotient map (continuous, open, onto) and π is constant on its fibres (G transitive on fibres), so π induces a continuous inverse $B \to X/G$. ■

Example 13.4

On the cylinder $X = S^1 \times I$, $h(x,t) = (-x,t)$ and $k(x,t) = (-x, 1-t)$ generate properly discontinuous $\mathbb{Z}/2$-actions; $X/\langle h \rangle$ is a cylinder, $X/\langle k \rangle$ the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ on S^n gives P^n.

13.4 Existence of covering spaces

Motivation. If B is semilocally simply connected, every subgroup of $\pi_1(B)$ comes from a covering.

B is **semilocally simply connected** if each b has a neighbourhood U with $\pi_1(U,b) \to \pi_1(B,b)$ trivial; this is necessary for a universal covering (Lemma 13.8).

Theorem 13.15 (82.1)

Let B be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and H a subgroup of $\pi_1(B,b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E,e_0) = H$.

Proof.

Step 1. Let $\mathcal{P}$ be the set of paths in B from b_0; $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar{\beta}] \in H$ (an equivalence relation). Write $\alpha^\#$ for the class, E for the set of classes, $p(\alpha^\#) = \alpha(1)$. Then:

(1) $[\alpha] = [\beta]$ implies $\alpha^\# = \beta^\#$;

(2) $\alpha^\# = \beta^\#$ implies $(\alpha * \delta)^\# = (\beta * \delta)^\#$ for any path δ from $\alpha(1)$, as $[(\alpha * \delta) * \overline{(\beta * \delta)}] = [\alpha * \bar{\beta}]$.

Step 2. For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood U of $\alpha(1)$ let $B(U,\alpha) = \{(\alpha * \delta)^\# : \delta$ a path in U from $\alpha(1)\} \ni \alpha^\#$. If $\beta^\# \in B(U,\alpha)$, then $B(U,\beta) = B(U,\alpha)$: with $\beta^\# = (\alpha * \delta)^\#$, $(\beta * \gamma)^\# = (\alpha * (\delta * \gamma))^\# \in B(U,\alpha)$ by (2), and $\alpha^\# = (\beta * \bar{\delta})^\#$ gives the reverse. Hence if $\beta^\# \in B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$ and $V \subset U_1 \cap U_2$ is a path-connected neighbourhood of $\beta(1)$, $B(V,\beta) \subset B(U_1,\beta) \cap B(U_2,\beta) = B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$. So these sets form a basis.

Step 3. p is continuous ($B(U,\alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U,\alpha)) = U$).

Step 4. Let U be a path-connected neighbourhood of b_1 with $\pi_1(U,b_1) \to \pi_1(B,b_1)$ trivial. $p^{-1}(U)$ is the union of the $B(U,\alpha)$, $\alpha(1) = b_1$: if $p(\beta^\#) \in U$, take δ in U from b_1 to $\beta(1)$ and $\alpha = \beta * \bar{\delta}$; then $\beta^\# = (\alpha * \delta)^\#$. Distinct such sets are disjoint (Step 2). p is injective on $B(U,\alpha)$: if $(\alpha * \delta_1)(1) = (\alpha * \delta_2)(1)$, $\delta_1 * \bar{\delta}_2$ is a loop in U at b_1, nulhomotopic in B, so $[\alpha * \delta_1] = [\alpha * \delta_2]$. As a continuous open bijection, $p|B(U,\alpha)$ is a homeomorphism onto U.

Step 5. Let e_0 be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, $\tilde{\alpha}(c) = \alpha_c^\#$; it lifts α from e_0 to $\alpha^\#$. Continuity at c: a basic neighbourhood of $\tilde{\alpha}(c)$ is $B(U,\alpha_c)$, U a path-connected neighbourhood of $\alpha(c)$; for d near c, $\alpha([c,d]) \subset U$ and $[\alpha_d] = [\alpha_c * \delta]$, δ the reparametrized restriction of α to $[c,d]$ (or $[d,c]$, reversed), so $\tilde{\alpha}(d) \in B(U,\alpha_c)$.

Step 6. E is path connected (Step 5) and locally path connected (locally homeomorphic to B), so p is a covering map.

Step 7. For a loop α at b_0, its lifting from e_0 ends at $\alpha^\#$, which is e_0 iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E,e_0) = H$. ∎

Corollary 13.16 (82.2)

B has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

Proof. Necessity: Lemma 13.8 (each point of B is the image of a point of E).

Sufficiency: Theorem 13.15 with H trivial. ∎

With Theorem 13.4: for such B, equivalence classes of coverings $\leftrightarrow$ conjugacy classes of subgroups of $\pi_1(B,b_0)$.

13.5 Selected exercises

Motivation. Applications of the lifting lemma to maps into S^1, and a criterion for proper discontinuity (Theorem 13.13).

Exercise 13.1 (§79, Ex. 1; maps of spheres into the circle)

Show that if $n > 1$, every continuous map $f : S^n \to S^1$ is nulhomotopic.

Solution.

Step 1. S^n is path connected, locally path connected (locally homeomorphic to $\mathbb{R}^n$ by stereographic projection), and simply connected for $n \geq 2$ (Theorem 9.40).

Step 2. Let $p : \mathbb{R} \to S^1$ be the covering of Theorem 9.9, $x_0 \in S^n$, $e_0 \in p^{-1}(f(x_0))$. $f_*(\pi_1(S^n, x_0))$ is trivial, so lies in $p_*(\pi_1(\mathbb{R}, e_0))$, and Lemma 13.1 gives $\tilde{f} : S^n \to \mathbb{R}$ with $p \circ \tilde{f} = f$.

Step 3. $H(x,t) = p((1-t)\tilde{f}(x))$ is a homotopy from f to the constant $p(0)$. ∎

Exercise 13.2 (§79, Ex. 2; maps into the circle)

(a) Show that every continuous map $f : P^2 \to S^1$ is nulhomotopic.
(b) Show that the projection $q : T \to S^1$ of the torus $T = S^1 \times S^1$ onto its first factor is not nulhomotopic.

Solution.

(a) P^2 is a surface (Theorem 9.43), so path connected and locally path connected. $\pi_1(P^2, x_0)$ has order 2 (Corollary 9.44), so f_* of it is a finite subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ (Theorem 9.16), hence $\{0\}$. Steps 2–3 of Exercise 13.1 apply with P^2 for S^n.

(b)

Step 1. Let $j : S^1 \to T$, $j(z) = (z, b_0)$; then $q \circ j = \mathrm{id}_{S^1}$.

Step 2. A homotopy H from q to a constant gives the homotopy $H \circ (j \times \mathrm{id}_I)$ from id_{S^1} to a constant.

Step 3. By Lemma 9.20, $(\mathrm{id}_{S^1})_*$ would be trivial, but it is the identity of $\pi_1(S^1, b_0) \cong \mathbb{Z} \neq 0$. So q is not nulhomotopic. ∎

Exercise 13.3 (§81, Ex. 4; free actions of finite groups)

Let X be Hausdorff and G a finite group of homeomorphisms of X acting fixed-point freely (no $g \neq e$ has a fixed point). Show that the action of G is properly discontinuous.

Solution. Let $G = \{g_0 = e, g_1, \ldots, g_k\}$ and $x \in X$.

Step 1. The $g_i(x)$ are distinct: $g_i(x) = g_j(x)$ gives $g_j^{-1} g_i(x) = x$, so $g_j^{-1} g_i = e$, $i = j$.

Step 2. For $i \neq j$ choose disjoint open $W_{ij} \ni g_i(x)$, $W_{ji} \ni g_j(x)$, and let $V_i = \bigcap_{j \neq i} W_{ij}$: open, pairwise disjoint, $g_i(x) \in V_i$.

Step 3. $U = \bigcap_{i=0}^{k} g_i^{-1}(V_i)$ is an open neighbourhood of x with $U \subset V_0$ and $g_i(U) \subset V_i$, so $g_i(U) \cap U = \emptyset$ for $g_i \neq e$. ∎